

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.C.A. (CBCS) Degree Examinations

January 2024

22MCA101 - DATA STRUCTURES WITH ALGORITHMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I

	Marks	BT*	CO*	PO*
1. a) How does Structure differ from an array? Create a structure to store rollno, name and total_marks of five students and display the information.	10	L*3	1	2
b) Define Stack. Explain the primitive operations that can be performed on stack and give the C implementation code of the operations.	10	L3	1	2
2. a) Give your views on ADT. Write an ADT for rational numbers.	10	L3	1	2
b) Demonstrate an algorithm to evaluate an expression given in postfix notation. Trace your algorithm for the following expression $AB^CD-EF/GH+/*$. A=2, B=2, C=6, D=2, E=12, F=2, G=2, H=1.	10	L3	1	2

Unit – II

1. a) Describe the efficiency of recursion. Write a recursive C program to find n^{th} Fibonacci number and trace it for $n=3$.	10	L3	2	2
b) List the difference between ordinary queue and circular queue. Write a C program to realize Circular queue.	10	L3	2	2
4. a) Describe the Towers of Hanoi Problem. Write the recursive C function to solve this problem and illustrate for $n=3$ disks.	10	L3	2	2
b) Define recursion. List down the differences between iterative and recursive approach with example.	10	L2	2	1

Unit – III

5. a) Write C implementation code for the following: i) Delete a node (whose info field is specified) from an ordered linked list. ii) Insert a node into an ordered linked list.	10	L3	3	2
b) List the advantages of a doubly linked list over a singly linked list. Write a C program to implement integer Queue using doubly linked list.	10	L3	3	2
6. a) Write C implementation code for the following: i) Delete all the occurrences of a given number from the singly linked list. ii) Insert a node into a singly linked list after a given node.	10	L3	3	2
b) List the advantages of circular linked list over singly linked list. Write a C program to implement integer stack using singly linked list.	10	L3	3	2

Unit – IV

7. a) With a neat diagram explain algorithm development process.	10	L2	4	1
b) Distinguish between (i) Algorithm and pseudo code. (ii) Big oh and omega notation	10	L2	4	1
8. a) Define Binary search tree. Write an algorithm to insert an item into an ordered binary search tree.	10	L3	4	2
b) Briefly explain the various methods of implementing binary trees in C.	10	L3	4	2

Unit – V

9. a) Write and explain Quick sort technique with example.	10	L1	5	2
b) Explain the concept of binary search technique with an example. Is it efficient than sequential search?	10	L3	5	2
10. a) Write Prim's minimum spanning tree algorithm. Explain with example.	10	L2	5	2
b) Develop an algorithm for Depth First Search in Graph. Trace it using Stack.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
